

Problem

A 1-MVA substation operates at full load at 0.7 power factor. It is desired to improve the power factor to 0.95 by installing capacitors. Assume that new substation and distribution facilities cost \$120 per kVA installed, and capacitors cost \$30 per kVA installed.

(a) Calculate the cost of capacitors needed.

(b) Find the savings in substation capacity released.

(c) Are capacitors economical for releasing the amount of substation capacity?

Step-by-step solution

Step 1 of 5

$$\cos \theta_1 = 0.7 \Rightarrow \theta_1 = 45.57^\circ$$

$$S_1 = 1 \text{ MVA} = 1000 \text{ KVA}$$

$$P_1 = S_1 \cos \theta_1 = 700 \text{ KW}$$

$$Q_1 = S_1 \sin \theta_1 = 714.14 \text{ KVAR}$$

For improved P f,

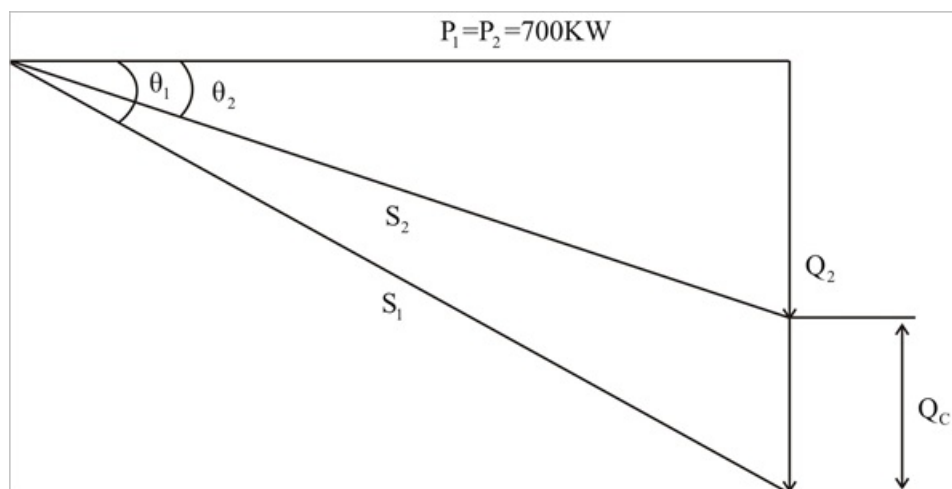
$$\cos \theta_2 = 0.95 \Rightarrow \theta_2 = 18.19^\circ$$

$$P_2 = P_1 = 700 \text{ KW}$$

$$S_2 = \frac{P_2}{\cos \theta_2} = \frac{700}{0.95} = 736.84 \text{ KVA}$$

$$Q_2 = S_2 \sin \theta_2 = 230.08 \text{ KVAR}$$

Step 2 of 5



Step 3 of 5

(A) Reactive power across the capacitor,

$$Q_c = Q_1 - Q_2 = 714.14 - 230.08 = 484.06 \text{ KVAR}$$

Cost of installing capacitor = \$ 30 × 484.06

= \$ 14,521.80 .

Step 4 of 5

(B) Substation capacity released = S1 - S2

$$= 1000 - 736.84 = 263.16 \text{ KVA}$$

Saving in cost of substation and distribution facilities,

$$= \$ 120 \times 263.16 = \$ 31,579.20$$

Step 5 of 5

(C) Yes, because (A) is greater than (B), additional system capacity obtained by using capacitors costs only 46 % as much as new substation and distribution faulties.